Assessment Menu: Teacher Guide

The Quadratics Bistro · design notes, grading and answer key

Why students cannot just pick the easy ones

1. **Points are capped by category.** Students must complete exactly 4 / 4 / 2 dishes, so they cannot fill the total with Appetizers.
2. **Chef's Specials carry 40% of the grade** and Entrées another 40%. A student who only masters the basics tops out at 20 points.
3. **Skill coverage rule.** Their 10 picks must include at least one dish from each skill group below, so every strand of the unit is assessed.
4. **Harder dishes reward process.** Each has a rubric that gives credit for correct steps, so a partly correct attempt at a harder problem earns more than a perfect easy one in the same time.

Skill group	What it covers	Dishes
FACTOR	Solve by factoring	A1, A5, E1, E2
ROOTS	Square roots, completing the square, formula and discriminant	A2, A4, E3, E4, S2
VERTEX	Vertex and axis of symmetry	A3, E5, S3
MODEL	Model projectile and area situations	A6, E6, S1, S4

If a student only does…	Maximum
Only Appetizers (4 × 5)	20
Appetizers + Entrées (4 × 5 + 4 × 10)	60
Full meal (adds 2 Chef's Specials)	100

Grading and options

- Grade only the marked dishes using the rubrics on the next pages (partial credit is built in). Total possible: **100**. If a student marks more than the required number, grade the first in each category.
- Check the skill rule on the order form before totaling. If a skill group is missing, apply your own policy (for example, the dish that broke the rule scores 0, or the student may swap one dish).
- Optional "second helping": allow one extra dish per category that replaces the student's lowest score in that category.
- Look at *which* dishes students chose. Students who select only the easiest dish in each skill group may need a conference.

Student	Appetizers (of 20)	Entrées (of 40)	Specials (of 40)	Skills OK?	Total

Answer Key: Appetizers (5 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
A1 FACTOR	Solve by factoring: $x^2 + 7x + 12 = 0$	**Answer:** $x = -3$ or $x = -4$ Find two numbers with product 12 and sum 7: 3 and 4 $(x + 3)(x + 4) = 0$ Set each factor to 0: $x = -3$ or $x = -4$	2 pts: Factors correctly as $(x + 3)(x + 4)$ 1 pts: Uses the zero product property 2 pts: Both solutions: $x = -3$ and $x = -4$
A2 ROOTS	Solve using square roots: $2x^2 - 50 = 0$	**Answer:** $x = 5$ or $x = -5$ Add 50: $2x^2 = 50$ Divide by 2: $x^2 = 25$ Take the square root of both sides: $x = \pm 5$	2 pts: Isolates $x^2 = 25$ 1 pts: Uses $\pm$ when taking the square root 2 pts: Both solutions: $x = 5$ and $x = -5$
A3 VERTEX	Find the axis of symmetry of the parabola $y = x^2 - 8x + 3$.	**Answer:** $x = 4$ Use $x = \frac{-b}{2a}$ with $a = 1$ and $b = -8$: $x = \frac{-(-8)}{2(1)} = \frac{8}{2}$ The axis of symmetry is $x = 4$.	2 pts: Identifies $a = 1$ and $b = -8$ 2 pts: Substitutes into the axis formula correctly 1 pts: Answer $x = 4$ written as an equation
A4 ROOTS	Use the discriminant to decide how many real solutions $x^2 + 4x + 7 = 0$ has. Show the value you find.	**Answer:** Discriminant $= -12$, which is negative, so there are no real solutions. Discriminant $= b^2 - 4ac$ with $a = 1$, $b = 4$, $c = 7$ $= 16 - 28 = -12$ A negative discriminant means **no real solutions**.	2 pts: Substitutes a, b, c correctly 1 pts: Discriminant $= -12$ 2 pts: Concludes no real solutions because it is negative
A5 FACTOR	Solve by factoring: $3x^2 - 12x = 0$	**Answer:** $x = 0$ or $x = 4$ Factor out the GCF $3x$: $3x(x - 4) = 0$ $3x = 0$ gives $x = 0$ $x - 4 = 0$ gives $x = 4$	2 pts: Factors out $3x$ 1 pts: Sets each factor equal to 0 2 pts: Both solutions: $x = 0$ and $x = 4$
A6 MODEL	A rectangle is 4 m longer than it is wide, and its area is 45 m². Write an equation for the width w, solve it, and give both dimensions.	**Answer:** $w(w + 4) = 45$; $w = 5$, so the rectangle is 5 m by 9 m. Length is $w + 4$, so $w(w + 4) = 45$ $w^2 + 4w - 45 = 0$, so $(w + 9)(w - 5) = 0$ $w = -9$ or $w = 5$; a width cannot be negative, so $w = 5$ Dimensions: 5 m by 9 m	2 pts: Correct equation $w(w + 4) = 45$ 1 pts: Factors correctly 1 pts: Rejects $w = -9$ 1 pts: States dimensions 5 m by 9 m

Answer Key: Entrées (10 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
E1 FACTOR	Solve by factoring: $3x^2 - 10x - 8 = 0$	**Answer:** $x = \frac{-2}{3}$ or $x = 4$ Find two numbers with product $3(-8) = -24$ and sum -10: -12 and 2 Split the middle term: $3x^2 - 12x + 2x - 8 = 0$ Group: $3x(x - 4) + 2(x - 4) = 0$, so $(3x + 2)(x - 4) = 0$ $x = \frac{-2}{3}$ or $x = 4$	3 pts: Splits the middle term correctly ($-12x$ and $2x$) 3 pts: Factors as $(3x + 2)(x - 4)$ 2 pts: Uses the zero product property 2 pts: Both solutions correct
E2 FACTOR	Ana solved $(x - 2)(x + 3) = 14$ by writing $x - 2 = 14$ or $x + 3 = 14$, so $x = 16$ or $x = 11$. Explain her mistake, then solve the equation correctly.	**Answer:** Zero product property needs the product to equal 0. Correct solutions: $x = -5$ or $x = 4$. The zero product property only works when the product equals **0**, not 14. Expand: $x^2 + x - 6 = 14$, so $x^2 + x - 20 = 0$ Factor: $(x + 5)(x - 4) = 0$ $x = -5$ or $x = 4$. Check $x = 4$: $(2)(7) = 14$ ✓	3 pts: Explains the product must be 0 before using the property 2 pts: Rewrites in standard form $x^2 + x - 20 = 0$ 2 pts: Factors correctly 3 pts: Both solutions: $x = -5$ and $x = 4$
E3 ROOTS	Solve by completing the square: $x^2 + 8x + 5 = 0$. Give exact answers.	**Answer:** $x = -4 \pm \sqrt{11}$ Move the constant: $x^2 + 8x = -5$ Add $(\frac{8}{2})^2 = 16$ to both sides: $x^2 + 8x + 16 = 11$ $(x + 4)^2 = 11$ $x + 4 = \pm\sqrt{11}$, so $x = -4 \pm \sqrt{11}$	2 pts: Moves the constant and adds 16 to both sides 2 pts: Writes $(x + 4)^2 = 11$ 2 pts: Takes the square root with $\pm$ 4 pts: Correct exact answers $x = -4 \pm \sqrt{11}$ (both signs)
E4 ROOTS	Solve using the quadratic formula: $2x^2 - 3x - 4 = 0$. Give exact answers.	**Answer:** $x = \frac{3 \pm \sqrt{41}}{4}$ $a = 2$, $b = -3$, $c = -4$ Discriminant: $(-3)^2 - 4(2)(-4) = 9 + 32 = 41$ $x = \frac{-(-3) \pm \sqrt{41}}{2(2)} = \frac{3 \pm \sqrt{41}}{4}$	2 pts: a, b, c with correct signs 3 pts: Discriminant = 41 2 pts: Substitutes correctly 3 pts: Correct exact answer
E5 VERTEX	For $y = -2x^2 + 8x - 3$, find the axis of symmetry and the vertex. Is the vertex a maximum or a minimum? Explain.	**Answer:** Axis $x = 2$; vertex $(2, 5)$; maximum because $a < 0$ (opens down). $x = \frac{-8}{2(-2)} = 2$; $y = -2(2)^2 + 8(2) - 3 = 5$ Vertex $(2, 5)$ $a < 0$, so it opens down: **maximum**.	3 pts: Axis of symmetry $x = 2$ 3 pts: Finds the y-coordinate 5 2 pts: Vertex written as $(2, 5)$ 2 pts: Maximum, with reason (a is negative)
E6 MODEL	A rectangle has perimeter 40 m and width w. Write its area A in terms of w. What width gives the greatest area, and what is that area?	**Answer:** $A = w(20 - w)$; greatest area 100 m² when $w = 10$ m (a square). Length $= 20 - w$, since $2w + 2(\text{length}) = 40$ $A = w(20 - w) = -w^2 + 20w$; vertex at $w = \frac{-20}{2(-1)} = 10$	3 pts: Length $= 20 - w$ 2 pts: $A = w(20 - w)$ 3 pts: Width 10 m (vertex) 2 pts: Greatest area 100 m² with units

		$A = 10(10) = 100$ Greatest area: 100 m^2	

Answer Key: Chef’s Specials (20 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
S1 MODEL	A rocket leaves a 24 ft platform. Its height is $h = -16t^2 + 40t + 24$ (feet, t in seconds). **(a)** How high is it at launch? **(b)** When does it hit the ground? Show how you decide which solution to use. **(c)** Find its maximum height and when it happens.	**Answer:** (a) 24 ft (b) t = 3 s (c) maximum 49 ft at t = 1.25 s (a) t = 0: h = 24 ft (b) Set h = 0: $-16t^2 + 40t + 24 = 0$. Divide by −8: $2t^2 - 5t - 3 = 0$ $(2t + 1)(t - 3) = 0$, so $t = \frac{-1}{2}$ or t = 3 Reject the negative time: **t = 3 s**. (c) $t = \frac{-40}{2(-16)} = 1.25$ s $h = -16(1.25)^2 + 40(1.25) + 24 = -25 + 50 + 24 =$ **49 ft**	2 pts: (a) 24 ft 2 pts: (b) Sets h = 0 3 pts: (b) Solves correctly 3 pts: (b) Rejects negative time; t = 3 s 3 pts: (c) t = 1.25 s 3 pts: (c) Height 49 ft 4 pts: Units and clear work
S2 ROOTS	Consider $x^2 + kx + 9 = 0$. **(a)** Find every value of *k* that gives exactly one real solution. **(b)** For the positive k from (a), solve the equation. **(c)** Explain, using the discriminant, why k = 4 gives no real solutions.	**Answer:** (a) k = 6 or k = −6 (b) x = −3 (c) discriminant = −20 < 0 (a) One solution means discriminant = 0: $k^2 - 4(1)(9) = 0$, so $k^2 = 36$ and **k = 6 or k = −6** (b) k = 6: $x^2 + 6x + 9 = 0$, so $(x + 3)^2 = 0$ and **x = −3** (c) k = 4: 16 − 36 = −20. A negative discriminant means no real square root, so **no real solutions**.	3 pts: Sets discriminant $k^2 - 36 = 0$ 3 pts: Both values k = 6 and k = −6 3 pts: (b) Factors $(x + 3)^2 = 0$ 2 pts: (b) x = −3 3 pts: (c) Computes discriminant −20 4 pts: (c) Explains that a negative discriminant means no real solutions 2 pts: Work clearly shown
S3 VERTEX	A parabola has vertex (3, −4) and passes through the point (1, 4). **(a)** Use $y = a(x - 3)^2 - 4$ to find a. **(b)** Write the equation in standard form. **(c)** Find the x-intercepts exactly.	**Answer:** (a) a = 2 (b) $y = 2x^2 - 12x + 14$ (c) $x = 3 \pm \sqrt{2}$ (a) Substitute (1, 4): $4 = a(1 - 3)^2 - 4$, so 8 = 4a and **a = 2** (b) $y = 2(x - 3)^2 - 4 = 2(x^2 - 6x + 9) - 4 =$ **$2x^2 - 12x + 14$** (c) Set y = 0: $2(x - 3)^2 = 4$, so $(x - 3)^2 = 2$ $x - 3 = \pm\sqrt{2}$, so **$x = 3 \pm \sqrt{2}$**	3 pts: (a) Substitutes (1, 4) correctly 3 pts: (a) a = 2 3 pts: (b) Expands $(x - 3)^2$ correctly 3 pts: (b) Correct standard form 4 pts: (c) Isolates $(x - 3)^2 = 2$ 4 pts: (c) Exact answers $3 \pm \sqrt{2}$
S4 MODEL	A 20 ft by 12 ft garden has a walkway of equal width *x* all the way around it. Garden and walkway together cover 660 ft². Sam wrote $(20 + x)(12 + x) = 660$. **(a)** Explain his error. **(b)** Write and solve the correct equation. **(c)** State the walkway width and the outer dimensions.	**Answer:** (a) Each dimension grows by 2x (b) $(20 + 2x)(12 + 2x) = 660$, x = 5 (c) 5 ft; 30 ft by 22 ft (a) The walkway is on **both** sides, so each dimension grows by 2x. (b) $(20 + 2x)(12 + 2x) = 660$, so $4x^2 + 64x + 240 = 660$ $4x^2 + 64x - 420 = 0$, then divide by 4: $x^2 + 16x - 105 = 0$ $(x + 21)(x - 5) = 0$, so x = −21 or x = 5; reject −21 (c) 5 ft wide; outer size 30 ft by 22 ft	4 pts: (a) Each dimension grows by 2x 4 pts: (b) Equation $(20 + 2x)(12 + 2x) = 660$ 4 pts: (b) Correct quadratic 3 pts: (b) Solves correctly 3 pts: (c) Rejects −21; 5 ft 2 pts: (c) Outer size 30 by 22